

EXCEL + AI | SUPPLY CHAIN ANALYTICS WORKSHOP

EXCEL + AI FOR SUPPLY CHAIN

DATA TO INSIGHTS DRIVEN DASHBOARD

From Raw Data to Actionable Business Insights

TRAINER

Dr Vinod Walwante

ORGANIZATION

Discover Talent

WORKSHOP DATE

30 / 08 / 2026

WHY EXCEL + AI FOR SUPPLY CHAIN?

THE CASE FOR FASTER, SMARTER ANALYSIS

Supply chain teams already have the data. The gap is the time it takes to turn it into a decision.

Too much operational data

Orders, stock, suppliers, deliveries in separate files

Manual reporting

Hours rebuilding the same report every week

Inventory visibility

No single view of what is short or sitting idle

Stock-outs

Lost sales and expedited freight

Excess inventory

Working capital locked in slow-moving items

Supplier delays

Late inbound pushes out the whole plan

Delivery performance

Promised vs actual dates tracked after the fact

Demand uncertainty

Plans built on last month, not on patterns

DATA



ANALYSIS



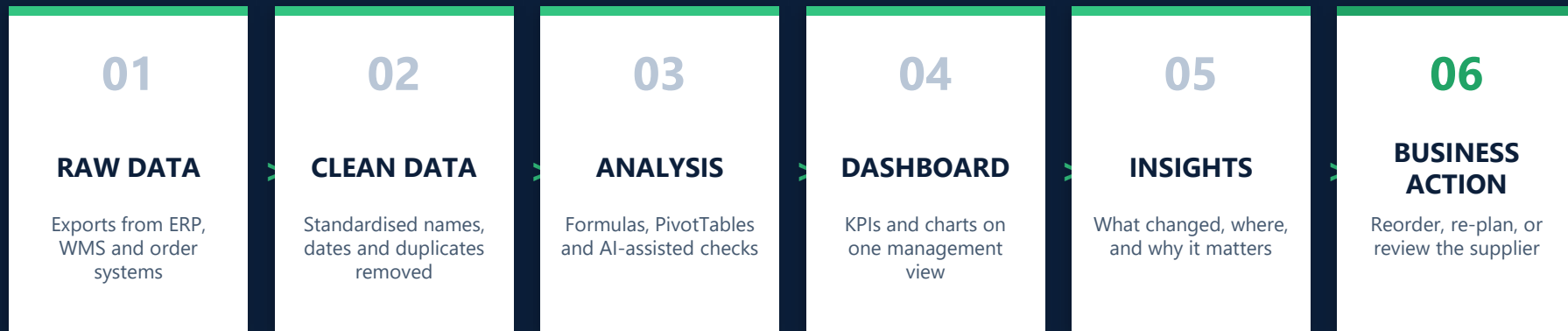
INSIGHTS



ACTION

FROM DATA TO DECISIONS

THE SIX-STAGE ANALYTICS FRAMEWORK



Each stage adds value. Skipping cleaning or insight turns the dashboard into decoration.

STAGE 1-2

GET THE DATA RIGHT

Standardise names, dates and duplicates before any calculation.

STAGE 3-4

MAKE IT MEASURABLE

Turn columns into KPIs, then into one management view.

STAGE 5-6

CLOSE THE LOOP

Explain the movement and assign the action and owner.

WHAT SUPPLY CHAIN DATA LOOKS LIKE

A REALISTIC ORDER-LEVEL DATASET

Order ID	Order Date	Product	Category	Supplier	Region	Warehouse	Qty	Sales	Stock	Reorder	Lead Time	Promised	Delivered
SO-1001	04-01-2026	Bearing 6204	Components	Apex Industrial	North	WH-Delhi	120	96,000	340	200	7	13-01-2026	12-01-2026
SO-1002	06-01-2026	Hydraulic Hose	Spares	Meridian Supply	West	WH-Pune	60	54,000	48	80	12	20-01-2026	24-01-2026
SO-1003	09-01-2026	Control Valve	Components	Northbridge Ltd	South	WH-Chennai	35	87,500	0	40	15	26-01-2026	29-01-2026
SO-1004	11-01-2026	Filter Cartridge	Consumables	Apex Industrial	East	WH-Kolkata	400	60,000	910	250	6	19-01-2026	19-01-2026
SO-1005	14-01-2026	Drive Belt	Spares	Vertex Traders	North	WH-Delhi	150	37,500	115	120	9	25-01-2026	25-01-2026

Every KPI in this workshop is derived from these columns — no extra system required.

ORDER FIELDS

Order ID, Date, Product, Category

NETWORK FIELDS

Supplier, Region, Warehouse

INVENTORY FIELDS

Stock, Reorder Level, Lead Time

DELIVERY FIELDS

Promised Date vs Delivery Date

THE REAL PROBLEM: MESSY DATA

BEFORE > AFTER

BEFORE — RAW EXPORT

Supplier	Product	Order Date	Qty
apex industrial	bearing 6204	04/01/2026	120
APEX INDUSTRIAL	Bearing-6204	2026-01-04	120
Meridian supply	hydraulic hose	06-01-26	(blank)
Northbridge Ltd.	Control valve	9 Jan 2026	35

AFTER — CLEANED

Supplier	Product	Order Date	Qty
Apex Industrial	Bearing 6204	04-01-2026	120
Meridian Supply	Hydraulic Hose	06-01-2026	60
Northbridge Ltd	Control Valve	09-01-2026	35
Vertex Traders	Drive Belt	14-01-2026	150

COMMON ISSUES

Extra spaces

Inconsistent capitalisation

Duplicate records

Missing values

Incorrect date formats

Inconsistent product names

Inconsistent supplier names

Analysis built on messy data produces confident, incorrect answers.

EXCEL FOR DATA CLEANING

FUNCTIONS AND TOOLS THAT DO THE WORK

TRIM

Removes extra spaces

=TRIM(" apex ") > "apex"

PROPER

Standardises name case

=PROPER("apex
industrial")

UPPER

Forces upper case

=UPPER("wh-delhi")

LOWER

Forces lower case

=LOWER("EMAIL@Co.COM")

CLEAN

Strips non-printing characters

=CLEAN(A2)

Remove Duplicates

Deletes repeated order or supplier rows

Find & Replace

Fixes recurring spelling variants at once

Text to Columns

Splits combined fields and repairs dates

Data Validation

Stops bad entries from being typed in

Power Query

Repeatable cleaning that refreshes each month

Clean once with Power Query, and every future refresh stays clean.

AI + EXCEL: THE NEW ANALYTICS WORKFLOW

WHERE AI HELPS AN EXCEL USER

BUILD

Formula creation

Describe the logic, get the formula

Formula explanation

Understand an inherited nested formula

Data cleaning assistance

Spot inconsistent names and formats

ANALYSE

Data analysis

Structure the questions worth asking

Pattern identification

Surface repeat delays and demand shifts

KPI suggestions

Propose the metrics that fit the data

COMMUNICATE

Trend analysis

Explain movement across months

Business insight generation

Turn a number into a "so what"

Summary creation

Draft the management commentary

AI assists the analyst — the analyst validates the result.

HOW TO WRITE BETTER AI PROMPTS

A SIX-PART FRAMEWORK

ROLE

Who the AI should be

CONTEXT

The business situation

DATA

Columns and period covered

TASK

What to produce

RULES

Limits and assumptions

OUTPUT

The exact format wanted

EXAMPLE PROMPT — SUPPLIER DELAY REVIEW

Role: Act as a supply chain analyst. **Context:** On-time delivery fell last quarter and we need to know why.

Data: Order-level table with Supplier, Order Date, Lead Time, Promised Date, Delivery Date. **Task:** Rank suppliers by delayed orders and average delay in days.

Rules: Use only the columns given, state any assumption, do not estimate missing values. **Output:** A five-row table plus three bullet observations.

Name the columns, not just the file

State the period explicitly

Always ask for assumptions to be listed

Validate the output before you use it

USE CASE: INVENTORY ANALYSIS

WHAT THE STOCK DATA SHOULD TELL YOU

Stock-out risk

Items at or heading to zero cover

Low stock

Approaching the reorder trigger

Reorder requirements

What to raise, and in what quantity

Healthy stock

Within the intended operating band

Excess inventory

Cover far beyond demand

Inventory value

Working capital held in stock

EXAMPLE KPI CARDS — ILLUSTRATIVE FIGURES FROM THE SAMPLE DATASET

1,248

SKUs tracked

37

Stock-out items

94

Low-stock items

112

Reorder now

1,005

Healthy SKUs

4.6 Cr

Inventory value

Figures shown are workshop examples from the sample dataset, not benchmarks.

USE CASE: STOCK RISK & REORDER

A CLASSIFICATION EVERY SKU CAN BE SCORED AGAINST

STOCK OUT

Current Stock = 0

Nothing available to ship today

REORDER

Current Stock $\leq$ Reorder Level

Raise the purchase order now

LOW STOCK

Current Stock $\leq 1.5 \times$ Reorder Level

Watch closely, plan the next order

HEALTHY

Otherwise

No action required this cycle

AUTOMATE IT WITH ONE NESTED FORMULA

```
=IFS(Stock=0,"Stock Out", Stock<=Reorder,"Reorder", Stock<=1.5*Reorder,"Low Stock", TRUE,"Healthy")
```

Copy it down the column and the entire catalogue is classified in seconds — then filter, count and chart each status.

The 1.5x multiplier is a starting rule of thumb — set it to match your own service level and lead times.

USE CASE: SUPPLIER PERFORMANCE

WHO IS ACTUALLY DELIVERING

Total Orders

Volume placed with each supplier

On-Time Delivery %

Share met on or before promise

Delayed Orders

Count arriving after promise

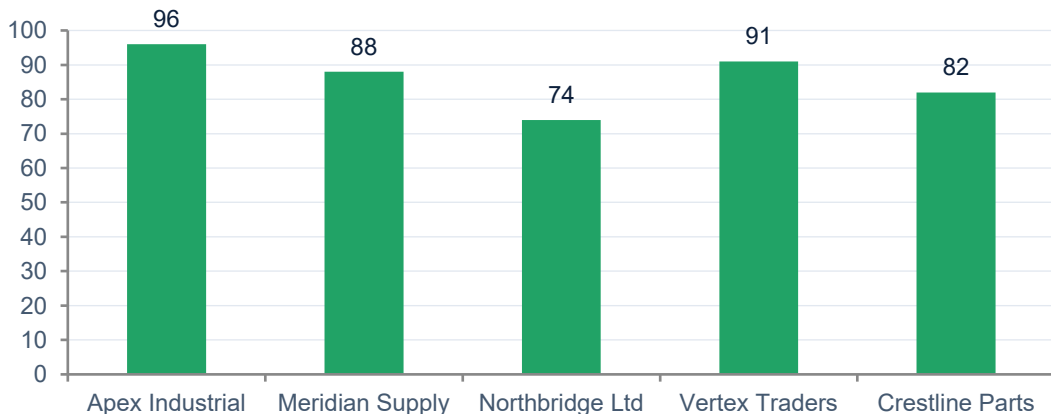
Average Lead Time

Order to receipt, in days

Supplier Performance

Combined service score

On-Time Delivery % by Supplier (sample data)



READING THE CHART

- Northbridge Ltd is the weakest performer at 74% on-time.
- Apex Industrial carries high volume and still holds 96%.
- Rank by delayed orders, not just percentage — volume matters.
- Pair the score with average lead time before renegotiating.

Values shown are illustrative workshop data used to demonstrate the calculation method.

USE CASE: DELIVERY PERFORMANCE

PROMISED DATE VS ACTUAL DELIVERY DATE

THE COMPARISON

Order	Promised	Delivered	Delay (days)	Status
SO-1001	13-01-2026	12-01-2026	-1	On Time
SO-1002	20-01-2026	24-01-2026	+4	Delayed
SO-1003	26-01-2026	29-01-2026	+3	Delayed
SO-1004	19-01-2026	19-01-2026	0	On Time
SO-1005	25-01-2026	25-01-2026	0	On Time

3 of 5

On-time deliveries

2 of 5

Delayed deliveries

1.4 days

Average delay

60%

Delivery performance

THE TWO FORMULAS THAT DRIVE IT

=IF(Delivery<=Promised,"On Time","Delayed")

=AVERAGEIF(Status,"Delayed",Delay)

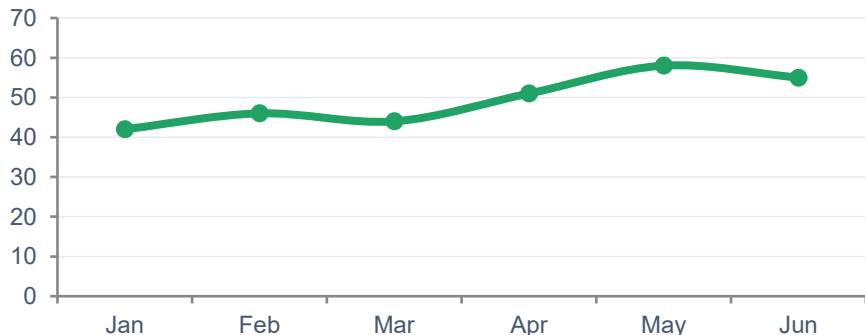
Add Supplier, Region or Warehouse to the PivotTable and the same measure explains where the delay sits.

Track the average delay alongside the percentage — a 60% score with a one-day slip is a different problem to a two-week slip.

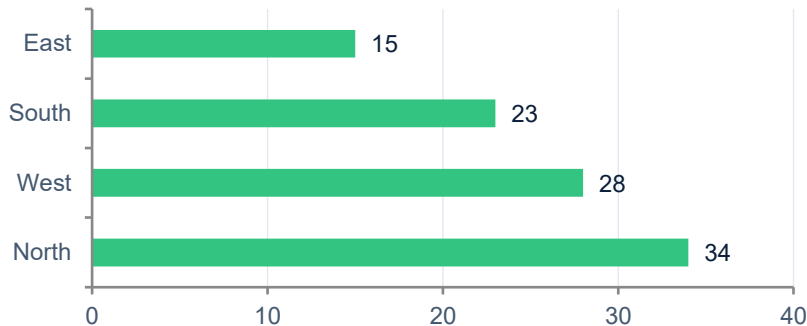
USE CASE: SALES & DEMAND INSIGHTS

WHAT IS SELLING, WHERE, AND WHEN

Monthly sales trend (sample data)



Regional sales share (sample data)



Top products

A small group drives most of the revenue

Low performers

Slow movers still consuming stock and space

Category performance

Which families are growing or fading

Regional performance

Where demand and service diverge

Monthly trends

Direction of travel, not a single month

Demand patterns

Seasonality that should shape reorder timing

WHAT SHOULD A SUPPLY CHAIN DASHBOARD SHOW?

KPIS THAT ANSWER A BUSINESS QUESTION

Total Sales

How much did we sell?

Total Orders

How much did we handle?

Inventory Value

How much capital is tied up?

Stock-out Items

What can we not supply today?

Low-stock Items

What is about to run short?

On-Time Delivery %

Are we keeping our promises?

Average Lead Time

How long from order to receipt?

Supplier Performance

Who needs a conversation?

Top Products

Where does the revenue come from?

Regional Performance

Where is demand strongest?

A KPI earns its place only if someone would act differently based on it.

Start from the question the business asks in the review meeting, then choose the metric — not the other way round.

Ten well-chosen KPIs beat thirty that nobody reads.

DASHBOARD DESIGN: FROM NUMBERS TO VISUALS

A MANAGEMENT-FRIENDLY WIREFRAME

KPI CARDS

Total Sales

Total Orders

Inventory Value

Stock-outs

On-Time %

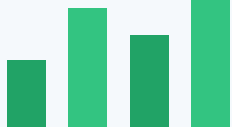
Avg Lead Time

CHARTS

Sales Trend



Inventory Status



Supplier Performance

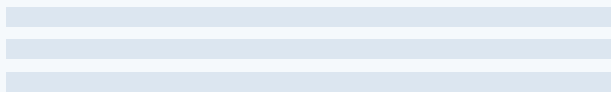


Delivery Performance

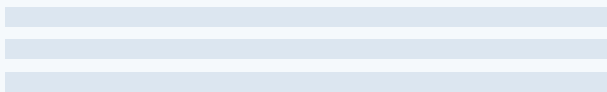


DETAIL

Product Analysis



Regional Analysis



FILTERS

Date

Region

Warehouse

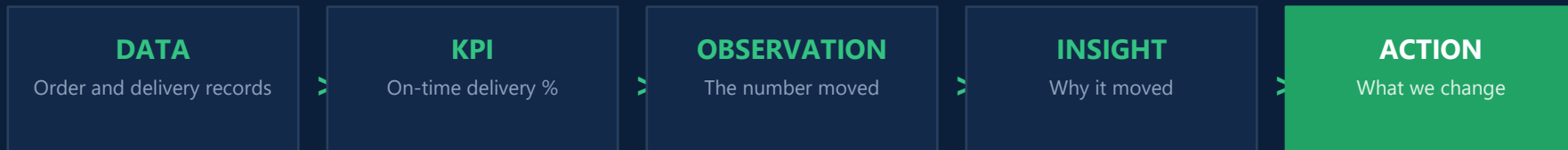
Category

Supplier

Top for the headline, middle for the trend, bottom for the detail — and filters where the eye lands last.

FROM DASHBOARD TO BUSINESS INSIGHTS

THE PROGRESSION THAT CREATES VALUE



WORKED EXAMPLE

OBSERVATION	On-time delivery decreased this quarter.
INSIGHT	Supplier A has the highest number of delayed orders.
ROOT CAUSE	Supplier A's average lead time increased.
ACTION	Review supplier performance and reorder timing.

A dashboard is not the final output. The decision is.

LIVE DEMO

RAW DATA > EXCEL + AI > DASHBOARD > INSIGHTS

1

Raw supply chain data

2

Clean the data

3

Apply Excel formulas

4

Use AI / Copilot

5

Calculate KPIs

6

Create PivotTables

7

Create PivotCharts

8

Build the dashboard

9

Generate insights

10

Recommend business
actions

RAW DATA

>

EXCEL + AI

>

DASHBOARD

>

INSIGHTS

Follow along in your own workbook — the instructor will walk through each step live.

EXCEL + AI FOR SUPPLY CHAIN

QUESTIONS?

Let's turn supply chain data into better decisions.

DATA

Clean it before you trust it

EXCEL

Formulas that classify and measure

AI

A faster assistant, always validated

DASHBOARD

KPIs that lead to a decision

Thank you for attending.